

Tasin Academy

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Marie Curie, née Maria Skłodowska, was born in Warsaw on November 7, 1867, the daughter of a secondary school teacher. From childhood, she was remarkable for her prodigious memory, and at the age of 16, she won a gold medal on completion of her secondary education at the Russian lycée. Because her father, a teacher of mathematics and physics, lost his savings through bad investments, she had to take work as a teacher and, at the same time, took part clandestinely in the nationalist "free university." From her earnings, she was able to finance her sister Bronisława's medical studies in Paris, with the understanding that Bronisława would in turn later help her to get an education.

In 1891, Skłodowska went to Paris and, now using the name Marie, began to follow the lectures of Paul Appell, Gabriel Lippmann, and Edmond Bouty at the Sorbonne. There she met many physicists who were already well known. Skłodowska worked far into the night in her student-quarter's garret and virtually lived on bread and butter and tea. She came first in the "licence examination" of physical sciences in 1893. She began to work in Lippmann's research laboratory and in 1894 was placed second in the licence examination of mathematical sciences. It was in the spring of that year that she met Pierre Curie.

Their marriage (July 25, 1895) marked the start of a partnership that was soon to achieve results of world significance, in particular the discovery of polonium (so called by Marie in honour of her native land) in the summer of 1898 and that of radium a few months later. Following Henri Becquerel's discovery (1896) of a new phenomenon (which she later called "radioactivity"), Marie Curie, looking for a subject for a thesis, decided to find out if the property discovered in uranium was to be found in other matter. She discovered that this was true for thorium at the same time as G.C. Schmidt did.

Turning her attention to minerals, she found her interest drawn to pitchblende, a mineral whose activity, superior to that of pure uranium, could be explained only by the presence in the ore of small quantities of an unknown substance of very high activity. Pierre Curie then joined her in the work that she had undertaken to resolve this problem and that led to the discovery of the new elements, polonium and radium. While Pierre Curie devoted himself chiefly to the physical study of the new radiations, Marie Curie struggled to obtain pure radium in the metallic state — achieved with the help of the chemist André-Louis Debierne, one of Pierre Curie's pupils. On the results of this research, Marie Curie received her doctorate of science in June 1903 and, with Pierre, was awarded the Davy Medal of the Royal Society. Also, in 1903, they shared with Becquerel the Nobel Prize for physics for the discovery of radioactivity.

The birth of her two daughters, Irène and Ève, in 1897 and 1904, did not interrupt Marie's intensive scientific work. She was appointed lecturer in physics at the École Normale Supérieure for girls in Sèvres (1900) and introduced there a method of teaching based on experimental demonstrations. In December 1904, she was appointed chief assistant in the laboratory directed by Pierre Curie.

1A. Multiple Choice Questions

1. The word 'prodigious' refers to _____.
(i) very weak (ii) extremely large or impressive (iii) ordinary (iv) unnoticeable
2. The word 'clandestinely' refers to something that is done _____.
(i) in secret (ii) openly (iii) quickly (iv) without care
3. Marie Curie's father lost his savings because of _____.
(i) political issues (ii) family disputes (iii) bad investment (iv) personal health expenses

4. **A physicist is _____.**
 (i) a scientist who studies physics (ii) a scientist who studies chemistry
 (iii) a historian (iv) a philosopher
5. **Marie Curie was basically _____.**
 (i) a scientist (ii) a poetess (iii) an engineer (iv) a physician
6. **Marie Curie supported her sister's education by _____.**
 (i) working as a teacher (ii) borrowing money from relatives
 (iii) receiving government aid (iv) joining a university
7. **Marie Curie's early interest in education is shown through her involvement in _____.**
 (i) the Russian lycée (ii) the "free university" for women
 (iii) university studies in Paris (iv) her father's school
8. **Marie Curie's remarkable talent as a student was evident when she _____.**
 (i) won a gold medal at 16 (ii) completed her PhD
 (iii) published scientific papers (iv) taught physics at a university
9. **What does the passage suggest about Marie Curie's financial situation in her youth?**
 (i) She struggled due to family financial problems. (ii) She had enough money from her teaching.
 (iii) She relied on scholarships. (iv) She received help from the government.
10. **Marie Curie struggled to abstain pure radium in the _____ state.**
 (i) floating (ii) waning (iii) burning (iv) metallic

1B. Short Questions

- (a) How did Marie Curie contribute to her sister's education, and what was the agreement between them?
- (b) How did Marie Curie's early academic achievements influence her future career?
- (c) Do you think Marie Curie's experience as a governess helped shape her determination to pursue higher education? Why or why not?
- (d) When did Sklodowska go to Paris and what did she begin to do there?
- (e) Why did she work so hard and what was her achievement?

2. Table Completion with the given information

What/Who	Events/Activity	Where/Place	When/How/Why
Marie Curie	Was born	(i)	(ii).....
(iii).....	Was a school teacher	(iv).....	
Marie Curie	Was remarkable		(v).....
(vi).....	Won a gold medal		(vii).....
She	Had to work as a teacher		(viii).....
She	(ix).....	In Russia	1985
She	Was able to finance her sister's education	(x).....	

Or,

Read the above passage and make a flowchart showing Marie Curie's family state. (One is done for you.) 2x5=10

1. Had her birth in Warsaw	→	2	→	3	→	4	→	5	→	6
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3. Summary Writing

Write a summary of the passage in your own words.